

LK VETRIINFRA

Complete Real Estate Plot Management Platform

User App (Customer) + Admin App (Agent / Business)

Manage | Grow | Close More Deals

Tech Stack	React (Vite) + Tailwind CSS + Firebase
Apps Included	Public Customer App + Internal Admin/Agent App
Cost Model	100% free-tier tools for build, hosting and analytics
Purpose	Full project blueprint an AI coding agent can build from directly

This document defines the complete idea, page-by-page design, Firebase data model, responsive layout rules, analytics plan and ready-to-use AI build prompts for the LK Vetriinfra platform. It is written so that the whole PDF (or any single section) can be pasted into an AI coding agent (Claude, ChatGPT, Cursor, etc.) to generate a production-quality React + Tailwind CSS + Firebase application, for both the public user-facing app and the internal admin/agent app.

1. Project Overview

LK Vetriinfra is a two-sided real estate plot management platform. It is made of **two connected applications** that share one Firebase backend:

- **Admin / Agent App** — internal tool for the business: manage plot inventory, customers/leads, appointments, sales & profit, documents and reports (this is the app shown in the reference screens).
- **Customer / User App** — a public-facing website + mobile-friendly app where buyers browse plots, save favourites, raise enquiries, book site visits and track their own journey.

Core Objectives

- Give the public a clean, trustworthy way to discover and enquire about plots online.
- Give the sales team one place to manage inventory, leads, appointments, sales and profit.
- Track every lead from first enquiry to closed sale (full funnel visibility).
- Automate price calculation, follow-up reminders and status transitions.
- Store plot, customer and transaction documents securely in the cloud.
- Provide high-quality analytics and reports for business decisions.
- Work perfectly from a 360px mobile screen up to a large desktop monitor.
- Use only free-tier / no-cost tools so the project can be run at zero infrastructure cost initially.

Who Uses What

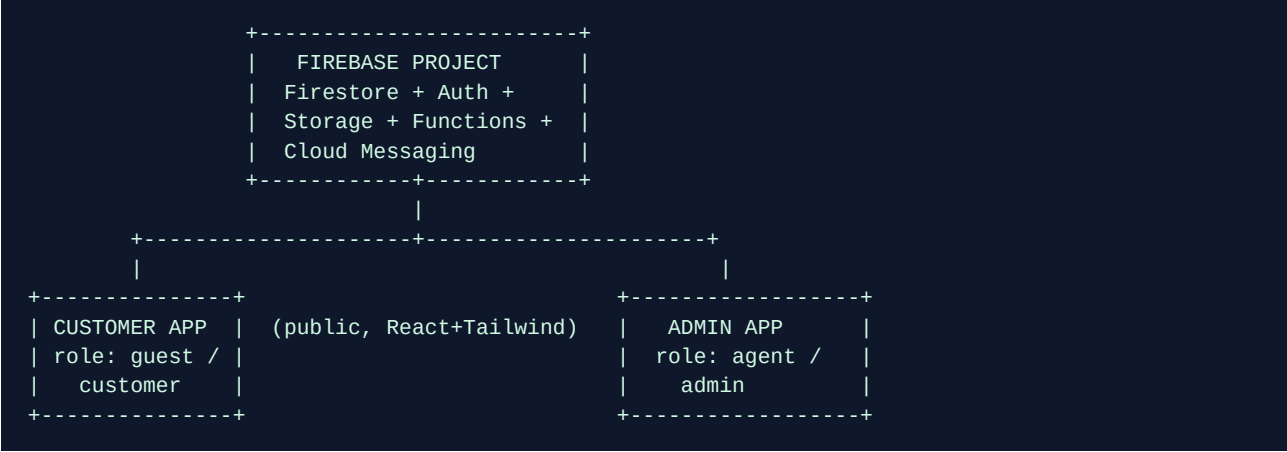
Role	App	What they do
Visitor / Buyer	Customer App	Browse plots, view details, use EMI calculator, submit enquiries
Registered Customer	Customer App	Save wishlist, book/track site visits, view offers, chat/WhatsApp agent
Sales Agent	Admin App	Manage assigned leads, appointments, plot updates, site visit notes
Admin / Owner	Admin App	Full access: plots, customers, sales, profit, reports, settings, users

LK Vetriinfra — Project Blueprint & AI Build Guide — Page will auto-number in the final PDF.

2. Two-Application Architecture

Both apps are built with the **same design system** and talk to the **same Firebase project**, so data entered by an agent (a new plot, a status update) is instantly visible to customers, and every customer enquiry instantly appears in the admin app.

Shared Firebase Backend (text diagram)



Why one backend, two front-ends

- No duplicate data entry — plots, prices and status live in one Firestore database.
- Firestore Security Rules (not app code) decide who can read/write what — see Section 9.
- Both apps can be deployed independently on Firebase Hosting under two sites, e.g. **app.lkvetriinfra.com** (admin) and **lkvetriinfra.com** (customer).
- A monorepo (Section 17) lets both apps share types, Firebase config and UI components.

3. Recommended Tech Stack (Free-First)

Layer	Technology	Free tier
Frontend	React (Vite) + Tailwind CSS	Fully open-source, no cost
Language	TypeScript (recommended)	Free
Routing	React Router	Free
Forms & Validation	React Hook Form + Zod	Free
Icons	Lucide React	Free
Charts / Analytics UI	Recharts	Free
Database	Firebase Firestore	Spark (free) plan: 1 GiB storage, 50k reads/day
Authentication	Firebase Authentication	Email/Password, Google, Phone OTP — free
File Storage	Firebase Storage	Free 5 GB (or Cloudinary free 25 GB for images/PDFs)
Hosting	Firebase Hosting	Free SSL + CDN + custom domain
Server Logic	Firebase Cloud Functions	Free quota for reminders, PDF/report generation
Push Notifications	Firebase Cloud Messaging	Free, unlimited
Email Sending	EmailJS or Firebase Extensions (Trigger Email)	Free tier available
WhatsApp Contact	wa.me click-to-chat links	100% free, no API needed
Maps	Leaflet + OpenStreetMap (or Google Maps Embed)	Free, no billing card required
Product Analytics	Firebase Analytics + Google Analytics 4	Free, unlimited events
PDF Export	jsPDF / react-pdf	Free, client-side
Design / Mockups	Figma (free plan)	Free
Version Control / CI	GitHub + GitHub Actions	Free for personal projects

Note: Every tool above has a genuinely free tier that is enough to launch and run this project for a small-to-mid size real-estate business. Upgrade only if traffic or storage genuinely outgrows the free quota.

4. Responsive Layout Specification

Mobile-first. The same information architecture adapts across breakpoints instead of building separate apps.

Viewport	Layout	Navigation	Main content
360–639px	Single column	Bottom navigation + FAB	Cards / stacked forms
640–767px	Single/two column	Bottom nav or compact rail	2-column cards where useful
768–1023px	Two-column	Sidebar or compact rail	Responsive grid
1024–1279px	Desktop	Full sidebar	2–3 column content
1280px+	Wide desktop	Full sidebar	Max-width content + 3–4 column grids

Mobile layout rules

- Fixed bottom navigation: Home, Plots/Browse, Add (admin) or Wishlist (customer), Customers/Enquiries, More.
- Large central circular action button (Add Plot on admin, Enquire Now on customer).
- Touch targets at least ~44px high; full-width cards with 12–16px page padding.
- One field per row in forms, except short related fields (e.g. Area + Rate).

Tablet layout rules

- Compact sidebar or adaptive navigation; 2-column grids for stat/property cards.
- Forms may use two columns for related fields; charts/tables stay scroll-free where possible.

Desktop layout rules

- Fixed left sidebar (240–280px); top header for search, notifications, profile.
- 3–4 column grids for summary cards; full data tables replace cards where more efficient.

5. Global Design System

Theme: modern, trustworthy, premium, real-estate focused — matching the LK Vetriinfra green + navy brand.

Token	Value	Usage
Primary Green	#16A34A	Main actions, active states, CTAs
Dark Green	#15803D	Headings, section bars, emphasis
Light Green	#DCFCE7	Success backgrounds, highlight chips
Navy / Text	#172033	Primary text, logo wordmark
Blue	#2563EB	Informational badges, links
Orange	#F97316	Appointments, attention states
Purple	#7C3AED	Analytics / reports accents
Red	#EF4444	Errors, destructive actions
Background	#F7F9FC	App background

Component rules

- Rounded cards, subtle borders, restrained shadows — no heavy skeuomorphism.
- Consistent status badges across plots, leads, appointments and payments (color-coded).
- Icon + label for every primary action; never an icon-only button for a critical action.
- Skeleton loading states for lists/dashboard cards; friendly empty states with a clear CTA.
- Confirmation dialogs before delete, mark-sold, cancel or any destructive action.
- Toast notifications after every successful create/update/delete.
- Customer app additionally needs: hero banners, trust badges, testimonials, and a warm, sales-friendly tone.

6. Complete Page Map — Admin / Agent App

Module	Pages
Authentication	Login, Forgot Password, Reset Password
Dashboard	Dashboard (KPIs, today's tasks, quick actions, recent activity)
Plots	Plot List, Add Plot, Plot Details, Edit Plot
Customers	Customer List, Add Customer, Customer Details, Edit Customer
Appointments	Appointment List, Add Appointment, Appointment Details
Sales	Sales & Profit Dashboard, Sale Details
Documents	Document List, Upload, Preview
Reports	Reports Dashboard, Sales, Plot, Customer, Profit, Lead Conversion, Agent Performance
Enquiries	Enquiry List (from Customer App), Enquiry Details, Convert to Customer
Messages	Conversation List, Conversation (customer chat / notes)
Settings	Profile, Company, Notifications, Users & Roles, Security, Appearance
More	Documents, Enquiries, Messages, Reports, Settings, About Us

7. Complete Page Map — Customer / User App

This is the new public-facing side of the platform, designed to turn website visitors into booked site visits.

Module	Pages
Authentication	Login / Sign Up (Email, Google, Phone OTP), Forgot Password
Home	Hero + search bar, Featured/Hot plots, Why Choose Us, Testimonials, CTA banner
Browse Plots	Plot Listing (filters: location, budget, area, status), Map view
Plot Details	Gallery, price, EMI calculator, location map, amenities, documents preview, 'Enquire Now' + 'Book Site Visit' buttons, similar plots
Search & Filters	Advanced filter drawer/page (price range, project, facing, road width)
Wishlist	Saved / shortlisted plots
Enquiry	Enquiry form (name, phone, budget, requirement, preferred plot)
My Appointments	Upcoming / completed site visits and meetings booked by the customer
My Profile	Edit profile, KYC documents upload, saved searches
EMI Calculator	Standalone loan/EMI calculator tool page
Notifications	Price drop, new project launch, appointment reminders
Reviews / Testimonials	Customer reviews and success stories
Contact / About Us	Company info, office locations map, phone/WhatsApp/email, social links
Support	FAQ, live chat / WhatsApp click-to-chat widget

8. Firebase Firestore Data Model

Suggested top-level Firestore collections shared by both apps (Firestore is schema-less; fields below are the recommended minimum).

Collection	Important fields
users	uid, name, email, phone, role (admin/agent/customer), photoURL, createdAt
projects	id, name, location, description, totalPlots, coverImage, createdAt
plots	id, projectId, plotNumber, surveyNumber, areaSqft, ratePerSqft, totalAmount, status (available/reserved/sold/blocked), facing, roadWidth, photos[], documents[], geo{lat,lng}
customers	id, name, phone, email, address, budget, interestedProjectId, interestedPlotId, status, nextFollowupDate, notes, assignedAgentId
appointments	id, customerId, plotId, type (site visit/meeting/call/registration/payment), date, time, status, notes
sales	id, plotId, customerId, saleAmount, cost, profit, saleDate, paymentStatus
documents	id, name, category, fileUrl, projectId, plotId, customerId, uploadedBy, createdAt
enquiries	id, customerName, phone, projectId, budget, source, status, createdAt (from Customer App)
wishlists	userId, plotIds[] (subcollection under users or its own collection)
reviews	id, customerId, plotId/projectId, rating, comment, createdAt
notifications	id, userId, title, message, type, isRead, createdAt

9. Roles, Authentication & Security Rules

Roles

Role	Access
guest	Customer App only — browse, view details, submit enquiry (no login required)
customer	Customer App, logged in — wishlist, my appointments, profile, notifications
agent	Admin App — only their assigned customers, appointments and plot updates
admin	Admin App — full access to all modules, users & roles, settings

Store the role as a field on the **users** document (or a Firebase custom claim set via a Cloud Function on user creation) and check it in Firestore Security Rules — never trust a role sent from the client.

Example Firestore Security Rules (starting point)

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {

    function isSignedIn() { return request.auth != null; }
    function role() {
      return get(/databases/{database}/documents/users/{request.auth.uid}).data.role;
    }

    match /plots/{plotId} {
      allow read: if true; // public catalog
      allow write: if isSignedIn() && role() in ['admin','agent'];
    }

    match /enquiries/{id} {
      allow create: if true; // anyone can enquire
      allow read, update: if isSignedIn() && role() in ['admin','agent'];
    }

    match /customers/{id} {
      allow read, write: if isSignedIn() && role() in ['admin','agent'];
    }

    match /users/{uid} {
      allow read, update: if isSignedIn() && request.auth.uid == uid;
      allow read: if isSignedIn() && role() == 'admin';
    }
  }
}
```

10. High-Analytics Plan

Analytics has two layers: **product analytics** (how people use the site — free via Firebase/GA4) and **business analytics** (sales, profit, conversion — built inside the Admin App with Recharts).

Events to track (Firebase Analytics + GA4 — free, unlimited)

- page_view, plot_view, plot_search, filter_used
- enquiry_submitted, site_visit_booked, wishlist_add / wishlist_remove
- signup, login, whatsapp_click, emi_calculator_used
- sale_completed, plot_status_changed (server-side, from Admin App)

Sales funnel to visualise on the Admin Dashboard

Lead-to-Sale Funnel

```
WEBSITE VISIT
  |
  v
PLOT VIEW / ENQUIRY -----> ENQUIRIES module
  |
  v
CUSTOMER CREATED -----> CUSTOMERS module (status: New)
  |
  v
SITE VISIT -----> APPOINTMENTS module
  |
  v
NEGOTIATION -> BOOKING -> PAYMENT
  |
  v
SALE COMPLETED -----> SALES & PROFIT module
  |
  v
REPORTS (conversion %, avg. days to close, agent performance)
```

Admin dashboard KPIs

- Total Plots, Available, Reserved, Sold, Total Customers, Appointments Today, Total Sales, Net Profit
- Monthly Sales vs Profit chart (Recharts bar/line)
- Lead source breakdown (Website, WhatsApp, Referral, Walk-in, Facebook/Instagram, Ads)
- Conversion rate: Enquiry → Site Visit → Booking → Sale
- Agent-wise performance (leads handled, conversions, revenue)

Reports module (exportable as PDF / Excel / CSV / Print)

- Sales Report, Plot Inventory Report, Customer Report, Appointment Report
- Profit Report, Lead Conversion Report, Agent Performance Report

11. AI Build Prompts — Admin / Agent App

Paste each prompt below into an AI coding agent one page/module at a time, or use the Master Prompt in Section 14 to generate everything at once.

Prompt A1 — App shell

Build the LK Vetriinfra ADMIN app shell using React (Vite) + Tailwind CSS + Firebase.
Mobile (360-639px): fixed bottom nav (Home, Plots, Add, Customers, More) + central Add FAB.
Tablet (640-1023px): compact sidebar, 2-column grids.
Desktop (1024px+): fixed left sidebar, top header, 3-4 column grids, full tables.
Create reusable: AppLayout, Sidebar, BottomNav, TopHeader, PageHeader, StatCard, StatusBadge, SearchBar, FilterButton, Modal, Toast, EmptyState, LoadingState, FAB.
Protect all routes with Firebase Auth + role check (admin/agent only).

Prompt A2 — Dashboard

Create the Admin Dashboard reading live counts from Firestore (plots, customers, appointments, sales).
KPI cards: Total Plots, Available, Reserved, Sold, Customers, Appointments, Total Sales, Net Profit.
Add Today's Tasks, Quick Actions (Add Plot/Customer/Appointment, View Reports) and a Recharts monthly Sales vs Profit chart. Responsive per the breakpoint spec in Section 4.

Prompt A3 — Plots module

Create Plot List, Add Plot, Edit Plot and Plot Details pages backed by the 'plots' Firestore collection.
List: search, filter, sort, tabs (All/Available/Reserved/Sold), responsive cards on mobile, table on desktop.
Add Plot fields: Project, Location, Survey Number, Plot Number, Area, Rate/Sq.ft, Status, Road Width, Facing, Description, Photos (Firestore Storage upload), Documents (PDF upload).
Auto-calculate Total Amount = Area x Rate/Sq.ft.

Prompt A4 — Customers module

Create Customer List, Add Customer, Customer Details pages backed by the 'customers' collection.
Fields: Name, Mobile, Email, Address, Budget, Interested Project/Plot, Deal Status, Next Follow-up, Notes.
Statuses: New, Interested, Site Visit, Negotiation, Booked, Follow-up, Converted, Lost.
Details page: profile, interested properties, appointments, notes, documents, activity timeline.

Prompt A5 — Appointments

Create Appointments pages (tabs: Upcoming, Completed, All) backed by 'appointments' collection.
Types: Site Visit, Office Meeting, Document Meeting, Follow-up Call, Registration, Payment.
Actions: Create, Edit, Reschedule, Complete, Cancel, Delete. Send a Firebase Cloud Messaging reminder notification the morning of the appointment via a scheduled Cloud Function.

Prompt A6 — Sales & Profit

Create Sales & Profit dashboard backed by 'sales' collection. KPI cards: Total Sales, Total Cost, Net Profit. Period selector: This Month/Quarter/Year/Custom. Recharts monthly bar chart for Sales vs Profit. Recent Sales list with Project, Plot, Customer, Amount, Cost, Profit, Date.

Prompt A7 — Documents

Create document management backed by Firebase Storage + 'documents' collection. Categories: Land Documents, Layout, Sale Agreement, Customer Documents, Receipts, Registration, Other.
Actions: Upload (with progress bar), Preview, Download, Share, Delete. Validate file type/size.

Prompt A8 — Reports

Create Reports & Analytics: Sales, Plot Inventory, Customer, Appointment, Profit, Lead Conversion, Agent Performance. Include date-range filters, KPI summaries, Recharts charts and data tables. Add export buttons: PDF (jsPDF), Excel/CSV, Print. Keep everything responsive and printable.

Prompt A9 — Enquiries & Messages

Create Enquiries (reads from 'enquiries' collection filled by the Customer App) with fields: Customer, Phone, Project, Budget, Requirement, Source, Date, Status. Actions: Call, WhatsApp, Assign to agent, Convert to Customer, Follow-up. Create Messages as a two-pane desktop / list-then-thread mobile layout for agent-customer notes.

Prompt A10 — Settings & Roles

Create Settings: Profile, Company Info, Notifications, Users & Roles (admin can invite agents and set role field on their user doc), Security (change password, 2FA), Appearance (Light/Dark/System) Mobile: grouped settings cards. Desktop: sidebar settings nav + content panel.

12. AI Build Prompts — Customer / User App

Prompt C1 — Public app shell & Home

Build the LK Vetriinfra CUSTOMER app shell (React + Vite + Tailwind + Firebase). Public, no login required to browse. Mobile: bottom nav (Home, Browse, Wishlist, Enquiries, More). Desktop: top nav. Home page: hero banner with search bar (location/budget), Featured/Hot plots carousel, 'Why Choose Us' section, testimonials, CTA banner ('Invest in Land, Invest in Your Future'), footer with contact/social links. Use the same green/navy branding as the Admin App.

Prompt C2 — Browse Plots & Filters

Create a Plot Listing page reading only 'available' + 'reserved' plots from Firestore (never expose internal cost/profit fields). Filters: location, project, budget range, area range, facing, status. Grid of plot cards (image, project, location, area, price, status badge). Add a map view toggle using Leaflet + OpenStreetMap markers.

Prompt C3 — Plot Details

Create a Plot Details page: photo gallery, price, area, facing, road width, description, embedded map, document previews (layout PDF), an EMI calculator widget, 'Enquire Now' and 'Book Site Visit' buttons that write to the 'enquiries'/'appointments' collections, a wishlist heart icon, and a 'Similar Plots' section.

Prompt C4 — Auth, Wishlist & Profile

Add Firebase Auth (Email/Password + Google + Phone OTP) sign up/login. Create My Wishlist (saved plots), My Appointments (site visits booked by this user, read from 'appointments' filtered by customerId), and My Profile (edit name/phone, upload KYC documents to Firebase Storage).

Prompt C5 — Enquiry, EMI Calculator & Support

Create a standalone Enquiry form (name, phone, budget, requirement, preferred project) that writes to 'enquiries'. Create a standalone EMI Calculator page (loan amount, interest rate, tenure -> monthly EMI, with a Recharts amortization chart). Add a floating WhatsApp click-to-chat button (wa.me link) on every page and an FAQ/Support page.

Prompt C6 — Notifications & Reviews

Add Firebase Cloud Messaging web push for logged-in customers (new project launch, price drop, appointment reminder). Create a Reviews/Testimonials page reading from 'reviews', plus a form for verified customers (status = Converted) to submit a review.

Prompt C7 — About / Contact

Create About Us (mission, vision, features) and Contact Us (office address with map, phone, WhatsApp, email, social links, Privacy Policy, Terms & Conditions). Use a professional marketing layout, fully responsive.

13. Step-by-Step Implementation Roadmap

Phase	Work
1	Firebase project setup (Auth, Firestore, Storage, Hosting) + React/Vite/Tailwind scaffolding for both apps
2	Design tokens, shared UI components, routing shell for Admin App
3	Firebase Authentication + protected routes + role field on users
4	Admin Dashboard with live Firestore counts
5	Plots module (list, add, edit, details) + Firebase Storage image/PDF upload
6	Customers module + follow-up workflow
7	Appointments module + Cloud Function reminders
8	Sales & Profit module + Recharts
9	Documents module
10	Reports module + PDF/Excel/CSV export
11	Enquiries + Messages modules
12	Settings, Users & Roles, About
13	Customer App shell + Home + Browse Plots + Plot Details
14	Customer Auth + Wishlist + My Appointments + Profile
15	Enquiry form, EMI calculator, WhatsApp button, Notifications, Reviews
16	Firestore Security Rules hardening + analytics event wiring (Firebase Analytics/GA4)
17	Responsive testing (360/390/430/768/1024/1280px) + bug fixing
18	Deploy both apps to Firebase Hosting + final QA

14. Final Master Prompt (Paste This to an AI Agent)

This single prompt summarises the entire project. Paste it as-is into an AI coding agent to generate a first complete build; then use the individual prompts in Sections 11–12 to refine each page.

MASTER PROMPT

Build a complete, production-ready platform called "LK VETRIINFRA" - Mobile Real Estate Plot Management System, made of TWO apps sharing one Firebase backend:

- 1) Public CUSTOMER app (browse plots, enquire, wishlist, book site visits)
- 2) Internal ADMIN app (manage plots, customers, appointments, sales, profit, reports)

STACK: React (Vite) + TypeScript + Tailwind CSS + React Router + Lucide React + Recharts + React Hook Form + Zod + Firebase (Auth, Firestore, Storage, Hosting, Cloud Functions, FCM).

BRAND: LK Vetriinfra | Tagline: Manage - Grow - Close More Deals

Colors: Primary Green #16A34A, Dark Green #15803D, Navy text #172033, Background #F7F9FC.

RESPONSIVE: 360-639px mobile with bottom nav + FAB; 640-1023px tablet with adaptive nav and 2-column grids; 1024px+ desktop with sidebar, top header and 3-4 column grids. No horizontal overflow anywhere.

ADMIN PAGES: Login, Dashboard, Plots (List/Add/Details/Edit), Customers (List/Add/Details/Edit), Appointments, Sales & Profit, Documents, Reports, Enquiries, Messages, Settings, About.

CUSTOMER PAGES: Login/Signup, Home, Browse Plots (+ map view), Plot Details (+ EMI calculator), Wishlist, Enquiry form, My Appointments, My Profile, Notifications, Reviews, Contact/About, FAQ.

DATA MODEL (Firestore): users, projects, plots, customers, appointments, sales, documents, enquiries, wishlists, reviews, notifications - fields as defined in Section 8 of this document.

AUTH & ROLES: Firebase Auth (Email/Password + Google + Phone OTP). Roles: guest, customer, agent, admin, stored on the user document and enforced via Firestore Security Rules (see Section 9) - never trust a client-supplied role.

ANALYTICS: Wire Firebase Analytics/GA4 events (plot_view, enquiry_submitted, site_visit_booked, wishlist_add, sale_completed). Build the lead-to-sale funnel and KPI dashboard described in Section 10 inside the Admin App using Recharts.

ENGINEERING: reusable components, TypeScript types for every entity, a services/ layer for all Firestore access, protected + role-based routes, client+security-rule validation, loading/empty/error/success states, toast notifications, confirmation dialogs before destructive actions, accessible controls, optimized/lazy-loaded images, code-split routes, zero dead buttons, zero TypeScript errors, zero avoidable console warnings.

Use realistic Indian real-estate sample data and INR currency. Make the final result look like a real commercial SaaS product, not a demo, and use only free-tier tools/services throughout.

15. Free Software & Tools Checklist

Need	Free tool to use
Code editor / AI pair-programmer	VS Code + Claude / Claude Code (free tier available)
Backend, DB, Auth, Storage, Hosting	Firebase Spark (free) plan
Design mockups	Figma (free plan)
Icons	Lucide React
Charts	Recharts
Maps (no billing card)	Leaflet + OpenStreetMap
Image optimisation	Squoosh (web, free) or Cloudinary free tier
Email sending	EmailJS free tier or Firebase 'Trigger Email' extension
Customer chat	WhatsApp click-to-chat (wa.me links) — no API cost
Version control	GitHub (free for personal/small team repos)
CI/CD	GitHub Actions (free minutes/month) + Firebase CLI deploy
Analytics	Firebase Analytics + Google Analytics 4 (both free, unlimited events)
Uptime monitor	UptimeRobot free plan
Domain (optional)	Firebase Hosting default subdomain is free; custom domain is the only paid item

16. Suggested Folder Structure (Monorepo)

Folder Structure

```
lk-vetriinfra/  
├── packages/  
│   ├── shared/           # shared TS types, Firebase config, UI kit  
│   │   ├── types/  
│   │   ├── firebase/  
│   │   └── components/  
│   ├── admin-app/       # Admin / Agent React app  
│   │   ├── src/  
│   │   │   ├── components/ (layout, common, forms, plots, customers, ...)  
│   │   │   ├── pages/      (dashboard, plots, customers, appointments, ...)  
│   │   │   ├── hooks/      services/ routes/ App.tsx  
│   │   └── customer-app/   # Public customer React app  
│   │       ├── src/  
│   │       │   ├── components/ (layout, plot-card, filters, emi-calculator, ...)  
│   │       │   ├── pages/      (home, browse, plot-details, wishlist, profile, ...)  
│   │       │   ├── hooks/      services/ routes/ App.tsx  
│   ├── functions/        # Firebase Cloud Functions (reminders, emails, reports)  
│   ├── firestore.rules  
│   ├── firebase.json  
│   └── README.md
```

17. Production & Responsive QA Checklist

- Test at 360px, 390px, 430px, 768px, 1024px, 1280px and larger widths.
- No horizontal page scrolling caused by tables, charts or long text.
- Responsive typography instead of fixed oversized headings.
- All primary actions reachable with one hand on mobile.
- Bottom navigation stays visible on mobile without overlapping content (safe-area padding).
- Wide tables convert to cards or controlled horizontal scroll on small screens.
- Dialogs become bottom sheets on mobile where appropriate.
- Upload controls are usable on touch devices; charts resize without clipping labels.
- Form values are preserved when validation fails; clear loading/empty/error states everywhere.
- Accessible labels, focus states and keyboard navigation.
- No dead navigation links or placeholder buttons that look functional but do nothing.
- TypeScript types for every Firestore entity; centralised services/ layer for all data access.
- Firestore Security Rules tested for guest, customer, agent and admin roles.
- Passwords never stored in plain text (Firebase Auth handles this natively).
- Images optimised and lazy-loaded; large modules code-split/lazy-routed.
- Console kept free of avoidable errors and warnings before every deploy.

LK Vetriinfra — Complete Project Blueprint. Format: A4 portrait, suitable for project documentation, academic submission, client presentation and as a direct build reference for AI coding agents.